

Thuy T. Le — Curriculum Vitae

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EDUCATION

University of North Carolina at Charlotte

Ph.D. in Applied Mathematics

Charlotte, NC, U.S.A.

August, 2019 - May, 2023

University of North Carolina at Charlotte

M.S. in Mathematical Finance

Charlotte, NC, U.S.A.

August, 2016 - May, 2018

National Economics University

B.S. in Mathematical Finance

Hanoi, Vietnam

August, 2008 - March, 2012

APPOINTMENTS

Fall 2023 - present

Postdoctoral Research Scholar

Department of Mathematics,
NC State University, North Carolina, U.S.A.

Fall 2019 - Spring 2023

Teaching Assistant & Research Assistant

Department of Mathematics and Statistics,
University of North Carolina at Charlotte, North Carolina, U.S.A.

Fall 2012 - Spring 2019

Lecturer

Department of Mathematics,
Banking Academy of Vietnam, Hanoi, Vietnam.

HONORS & AWARDS

- **Key Participant, National Science Foundation**, Grant No. DMS-2208159, The Carleman Contraction Principle for Three-Dimensional Phased and Phaseless Inverse Scattering, PI. Loc Nguyen, \$246,641
- **Graduate Student Travel Grant** to Spring 2023 AMS Sectional meetings, American Mathematical Society, U.S.A., 2023
- **Graduate Travel Grant**, University of North Carolina at Charlotte, Charlotte, NC, U.S.A., 2023
- **Graduate School Summer Fellowship**, University of North Carolina at Charlotte, NC, U.S.A., 2022
- **Graduate Travel Grant**, University of North Carolina at Charlotte, Charlotte, NC, U.S.A., 2021
- **Scholarship for Outstanding Performance** in the Master of Mathematical Finance Program, University of North Carolina at Charlotte, Charlotte, NC, U.S.A., 2018
- **The Vietnamese Government Full Scholarship** - Project 599 for Training Young Leaders Abroad, 2016-2018
- **Second Prize** in “Young Scientific Talents of Vietnam”, National Economics University, Hanoi, Vietnam, 2012
- **Excellent Student Scholarship**, National Economics University, Hanoi, Vietnam, 2011
- **First Prize** in “Olympiad in Informatics for Undergraduates”, National Economics University, Hanoi, Vietnam, 2009

RESEARCH INTERESTS

Inverse Problems, Numerical Methods, Numerical Analysis, Nonlinear Partial Differential Equations, Bayesian Optimization, and Machine Learning.

PUBLICATIONS

Journal Papers.....

1. Cong B. Van, **Thuy T. Le**, and Loc H. Nguyen, *The inverse initial data problem for anisotropic Navier-Stokes equations via Legendre time reduction method*, Communications in Nonlinear Science and Numerical Simulation, 161(2026), 110074.
2. **Thuy T. Le**, Cong B. Van, Trong D. Dang, and Loc H. Nguyen, *Inverse initial data reconstruction for Maxwell's equations via time-dimensional reduction method*, Journal of Computational Physics, 559 (2026), 114896.
3. **Thuy T. Le**, Phuong M. Nguyen, and Loc H. Nguyen, *Inverse scattering without phase: Carleman convexification and phase retrieval via the Wentzel-Kramers-Brillouin approximation*, Computer Methods in Applied Mechanics and Engineering, 448 (2026), 118439.
4. R. Lawrence Ives, Michael Read, Jeff Neilson, Thuc Bui, David Marsden, **Thuy T. Le** and Hien T. Tran, *Gyrotrons for Fusion Power Plants*, Fusion Science and Technology, 1-11, 2025
5. Ray Abney, **Thuy T. Le**, Loc H. Nguyen, and Cam Peters, *A Carleman-Picard approach for reconstructing zero-order coefficients in parabolic equations with limited data*, Applied Mathematics and Computation, 494, 129286, 2025
6. **Thuy T. Le**, Linh V. Nguyen, Loc H. Nguyen, and Hyunha Park, *The time dimensional reduction method to determine the initial conditions without the knowledge of damping coefficients*, Computers and Mathematics with Applications, 166, 77-90, 2024
7. Huynh P. N. Le, **Thuy T. Le**, Loc H. Nguyen, *The Carleman convexification method for Hamilton-Jacobi equations*, Computers and Mathematics with Applications, 59:173–185, 2024
8. Anuj Abhishek, **Thuy T. Le**, Loc H. Nguyen, Taufiqar Khan, *The Carleman- Newton method to globally reconstruct a source term for nonlinear parabolic equation*, Journal of Computational and Applied Mathematics, 445:115827, 2024
9. Dinh-Nho Hào, **Thuy T. Le** and Loc H. Nguyen, *The dimensional reduction method for solving a nonlinear inverse heat conduction problem with limited boundary data*, Communications in Nonlinear Science and Numerical Simulation, 128:107679, 2024
10. Phuong M. Nguyen, **Thuy T. Le**, Loc H. Nguyen, and Michael V. Klibanov, *Numerical differentiation by the polynomial-exponential basis*, Journal of Applied and Industrial Mathematics, 17, 928–942, 2023
11. **Thuy T. Le**, Vo A. Khoa, Michael V. Klibanov, Loc H. Nguyen, Grant Bidney, and Vasily Astratov, *Numerical verification of the convexification method for a frequency-dependent inverse scattering problem with experimental data*, Journal of Applied and Industrial Mathematics, 17, 908–927, 2023
12. **T. T. Le**, *Global reconstruction of initial conditions of nonlinear parabolic equations via the Carleman-contraction method*, Advances in Inverse Problems for Partial Differential Equations, volume 784, pages 145-167, Contemporary Mathematics, American Mathematical Society, 2023.
13. **T. T. Le**, L. H. Nguyen and H. V. Tran, *A Carleman-based numerical method for quasilinear elliptic equations with over-determined boundary data and applications*, Computers and Mathematics with

Applications, Volume 125, Pages 13-24, 2022

14. **T. T. Le**, L. H. Nguyen, *The gradient descent method for the convexification to solve boundary value problems of quasi-linear PDEs and a coefficient inverse problem*, Journal of Scientific Computing 91:74, 2022
15. **T. T. Le**, M. V. Klibanov, L. H. Nguyen, A. Sullivan and L. Nguyen, *Carleman contraction mapping for a 1D inverse scattering problem with experimental time-dependent data*, Inverse Problems, 38, 045002, 2022
16. M. Hashemitaheri, **T.T. Le**, H. Cherukuri, T. Khan *A Multivariate Newton-Raphson Method Approach to Extract Structural Dynamics Parameters During Milling Operations*, AeroMat Conference and Exposition: AeroMat 2022 - Pasadena Convention Center, Pasadena, CA, USA, 2022
17. M. V. Klibanov, **T. T. Le**, L. H. Nguyen, A. Sullivan, and L. Nguyen, *Convexification-based globally convergent numerical method for a 1D coefficient inverse problem with experimental data*, Inverse Problems and Imaging, 16, p. 1579-1618, 2022
18. **T. T. Le**, and L. H. Nguyen, *A convergent numerical method to recover the initial condition of nonlinear parabolic equations from lateral Cauchy data*, Journal of Inverse and Ill-posed Problems, 30, p 256-286, 2022
19. **T. T. Le**, L. H. Nguyen, T-P. Nguyen and W. Powell, *The quasi-reversibility method to numerically solve an inverse source problem for hyperbolic equations*, Journal of Scientific Computing 87:90, 2021
20. M. V. Klibanov, **T. T. Le** and L. H. Nguyen, *Convergent numerical method for a linearized travel time tomography problem with incomplete data*, SIAM Journal on Scientific Computing 42(2020) B1173-B1192, 2020

Preprint(s).....

1. Zixuan Ma, Chenfeng Huang, **Thuy T. Le**, Hien Tran, *Safe-handover causal bayesian optimization with dynamic expert trust*, submitted, 2025
2. **Thuy T. Le**, Minh-Binh Tran, and Loc H. Nguyen, *A globally convergent Carleman-Picard method for an inverse initial-value problem for a nonlinear diffusive coagulation-fragmentation equation*, arXiv:2603.21185, 2026.

In Preparation.....

1. **Thuy T. Le**, *The Fourier-Operation networks for the inverse scattering problem with limited back scattering data*, 2026
2. **Thuy T. Le** and Hien T. Tran, *Efficient quantification of vaccination effects using a Biologically-informed neural network*, 2026
3. **Thuy. T. Le** and Alexandria Traynham, *A globally convergent numerical method for solving a coefficient inverse problem in parabolic equations with lateral Cauchy data*, 2026

TEACHING

North Carolina State University

Instructor, Department of Mathematics

Raleigh, NC, U.S.A.

Fall 2023 - present

- Fall 2025: Precalculus I - MA 107 - 005 (100 students), Precalculus I - MA 107 - 006 (100 students),
- Spring 2025: Precalculus I - MA 107 - 003 (100 students)
- Fall 2024: Precalculus I - MA 107 - 005 (101 students), Precalculus I - MA 107 - 006 (100 students)

- Spring 2024: Elements of Calculus – MA 121 – 002 (240 students)
- Fall 2023: Elements of Calculus – MA 121 – 001 (247 students)

University of North Carolina at Charlotte

Instructor, Department of Mathematics and Statistics

Charlotte, NC, U.S.A.

Fall 2020 - Spring 2023

- Fall 2021: Calculus 1 - MATH 1241-006
- Spring 2021: Calculus 1 - MATH 1241-008, Differential Equations - MATH 2171-002
- Fall 2020: Calculus 1 - MATH 1241-091

Teaching Assistant, Department of Mathematics and Statistics

Fall 2019 - Spring 2023

- Spring 2022: Differential Equations - MATH 2171-500, Advanced Calculus of One Variable - MATH 3141-001
- Summer 2021: Differential Equations - MATH 2171-080, Differential Equations - MATH 2171-082
- Summer 2020: Differential Equations - MATH 2171-080
- Spring 2020: Differential Equations - MATH 2171-090, Advanced Calculus of One Variable - MATH 3141-001
- Fall 2019: Matrices & Linear Algebra - MATH 2164-008

Banking Academy of Vietnam

Instructor, Department of Mathematics

Hanoi Vietnam

Fall 2012 - Spring 2019

Taught courses:

- Theory of Probability and Statistics
- Econometrics
- Mathematics for Economists

MENTORSHIP

1. Chenfeng Huang, University of Southern California, Graduate Student
Submitted Paper: Zixuan Ma, Chenfeng Huang, Thuy T. Le, Hien Tran, *Safe-handover causal bayesian optimization with dynamic expert trust*, 2025
2. Cong Van, University of North Carolina at Charlotte, Graduate Student
Submitted Paper:
 - Cong B. Van, Thuy T. Le, and Loc H. Nguyen, *The inverse initial data problem for anisotropic Navier-Stokes equations via Legendre time reduction method*, arXiv:2507.16810, 2025
 - Thuy T. Le, Cong B. Van, Trong D. Dang, Loc H. Nguyen, *Inverse initial data reconstruction for Maxwell's equations via time-dimensional reduction method*, arXiv:2506.20777, 2025
3. Ray Abney, UNC Charlotte, Graduate Student
 Cam Peters, UNC Charlotte, Undergraduate Student
REU Program, NSF DMS-2150179
Published Paper: Ray Abney, Thuy T. Le, Loc H. Nguyen, and Cam Peters, *A Carleman-Picard approach for reconstructing zero-order coefficients in parabolic equations with limited data*, Applied Mathematics and Computation, 494, 129286, 2025
4. Phuong Nguyen, UNC Charlotte, Graduate Student
Published Paper: Phuong M. Nguyen, Thuy T. Le, Loc H. Nguyen, and Michael V. Klibanov, *Numerical differentiation by the polynomial-exponential basis*, Journal of Applied and Industrial Mathematics, 17, 928–942, 2023

5. Huynh Le, International Centre for Theoretical Physics, Trieste, Italy, Graduate Student
Published Paper: Huynh P. N. Le, Thuy T. Le, Loc H. Nguyen, *The Carleman convexification method for Hamilton-Jacobi equations*, Computers and Mathematics with Applications, 59:173–185, 2024
6. Hyunha Park, George Washington University, Undergraduate Student
REU Program, NSF DMS-2150179
Published Paper Thuy T. Le, Linh V. Nguyen, Loc H. Nguyen, and Hyunha Park, *The time dimensional reduction method to determine the initial conditions without the knowledge of damping coefficients*, Computers and Mathematics with Applications, 166, 77-90, 2024
7. William Powell, UNC Charlotte, Undergraduate Student
Published Paper: T. T. Le, L. H. Nguyen, T-P. Nguyen and W. Powell, *The quasi-reversibility method to numerically solve an inverse source problem for hyperbolic equations*, Journal of Scientific Computing 87:90, 2021
8. Alexandria Traynham, UNC Charlotte, Graduate Student
In-preparation Paper: Thuy. T. Le and Alexandria Traynham, *A globally convergent numerical method for solving a coefficient inverse problem in parabolic equations with lateral Cauchy data*

CONFERENCES AND SEMINARS

Seminar and Colloquium Talks.....

1. *An inverse scattering problem with experimental data*, Fluid Mechanics and Waves Seminar, New Jersey Institute of Technology, Newark, NJ, 2024
2. *The convexification method for systems of quasi-linear partial differential equations and its application to a coefficient inverse problem*, Applied Math Seminar, Kansas State University, Manhattan, Kansas State, 2020

Conference and Workshop Talks.....

1. *The dimensional reduction method for solving a nonlinear inverse heat conduction problem with limited boundary data*, 3rd UNCG Virtual PDE Conference, University of North Carolina at Greensboro, Greensboro, NC, 2025
2. *Carleman-Picard approach for a coefficient inverse problem in parabolic equations using partial boundary data*, AMS 2025 Spring Southeastern Sectional Meeting, Clemson University, Clemson, SC, 2025
3. *A convexification approach for the 3D inverse scattering problem with experimental data*, AMS 2024 Spring Southeastern Sectional Meeting, Florida State University, Tallahassee, FL, 2024
4. *A Global Convergent Numerical Method Based on A Carleman Estimate and The Contraction Mapping Principle for Solving An Inverse Scattering Problem in The Time Domain with Experimental Data*, SEARCDE 2023, Florida A&M University, Tallahassee, FL, 2023
5. *A Carleman-based numerical method for solving a 3D coefficient inverse problem*, AMS 2023 Fall Eastern Sectional Meeting, University at Buffalo (SUNY), Buffalo, NY, 2023
6. *A globally convergent numerical method for solving a coefficient inverse problem for a hyperbolic equation with experimental data*, 2023 UNC Greensboro PDE Conference, Greensboro, NC, 2023
7. *Global reconstruction of initial conditions for nonlinear parabolic equations via the Carleman-contraction method*, MS3: Recent Advances Differential Equations and Applications, SIAM Southeastern Atlantic Section Annual Meeting, Virginia Tech, Blacksburg, VA, 2023

8. *Convexification-based globally convergent numerical method for a 3D coefficient inverse problem*, Special Session on Recent Developments on Analysis and Computation for Inverse Problems for PDEs, AMS Spring Southeastern Sectional Meeting, Georgia Institute of Technology, Atlanta, GA, 2023
9. *Convexification method for the 3D inverse scattering problem in the frequency domain*, Third Conference on Analysis and Applied Mathematics (CAAM 3), Saigon University, Ho Chi Minh City, Vietnam, 2022
10. *The Carleman contraction mapping method for the 1D inverse scattering problem in the time domain with experimental data*, International Conference on Differential Equations and Applications, Hanoi, Vietnam, 2022
11. *Carleman Contraction Mapping for a 1D Inverse Scattering Problem with Experimental Time-Dependent Data*, MS137 - Recent Developments in Inverse Problems for Partial Differential Equations, SIAM Conference on Imaging Science (IS22), Virtual Conference, 2022
12. *A new Carleman estimate and the contraction principle for the 1D inverse scattering problem with experimental data*, Workshop on PDE and related topics, VIASM, Hanoi, Vietnam, 2022
13. *A Carleman-based reconstruction method for a 1D coefficient inverse problem with time-dependent experimental data*, Special Session on Recent Advances in Inverse Problems for PDEs II, AMS Fall Western Sectional Meeting, Virtual Conference, 2021
14. *Convexification-based globally convergent numerical method for a 1D coefficient inverse problem with experimental data*, MS6 - Recent Developments on Partial Differential Equations and Applications, SIAM Southeastern Atlantic Section Conference, Auburn University, Auburn, Alabama, 2021
15. *Reconstructing the initial condition of quasi-linear parabolic equations from lateral Cauchy data*, SS 3A – Special Session on Recent Developments on Analysis and Computation for Inverse Problems for PDEs, AMS Spring Southeastern Virtual Sectional Meeting, 2021
16. *A convergent numerical method to reconstruct the initial condition of nonlinear parabolic equations from lateral Cauchy data*, The October Math Day Symposium, University of North Carolina at Charlotte, Charlotte, North Carolina, 2020
17. *A convergent numerical method to reconstruct the initial condition of nonlinear parabolic equations from lateral Cauchy data*, The 15th UNCG Regional Mathematics and Statistics Conference, Greensboro, North Carolina, 2019

SERVICES

Conference Organizer.....

- Special Session on *Modeling and Control of Dynamical Systems and its Applications*, 2025 Spring Southeastern Sectional Meeting, Clemson University, Clemson, SC, March 8-9, 2025
- Special Session on *Recent Advances in Inverse Problems for Partial Differential Equations and Their Applications*, 2024 Spring Southeastern Sectional Meeting, Florida State University, Tallahassee, FL, March 23-24, 2024

Reviewer.....

- Journal of Computational Physics
- Evolution Equations and Control Theory
- Communications in Nonlinear Science and Numerical Simulation
- Scientific Reports

- AMS Contemporary Mathematics (CONM)
- BIT Numerical Mathematics
- AMS

REFERENCES

- **Prof. Loc Nguyen** (lnguye50@charlotte.edu)
Department of Mathematics and Statistics
University of North Carolina at Charlotte, Charlotte, NC, U.S.A
- **Prof. Hien Tran** (tran@ncsu.edu)
Department of Mathematics
Associate Department Head, Director for CRSC, Director of Multimedia Center
North Carolina State University, Raleigh, NC, U.S.A
- **Prof. Taufiqar Khan** (taufiqar.khan@charlotte.edu)
Chair of Department of Mathematics and Statistics
University of North Carolina at Charlotte, Charlotte, NC, U.S.A
- **Prof. Tien Khai Nguyen** (tnguye13@ncsu.edu)
Department of Mathematics
North Carolina State University, Raleigh, NC, U.S.A
- **Prof. Hung Vinh Tran** (hung@math.wisc.edu)
Department of Mathematics
University of Wisconsin Madison, WI, U.S.A